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course/program :- BTech C.S.E SY
Subject:- COA

COA Theory assignment 2.

Q.1) Describe the concept of memory hierarchy, with neat and clean diagram.

Sol:-

To achieve greatest performance, the memory must be able to keep up with the processor.

This is, as the processor is executing instructions, we would not went it to have to pause waiting for instructions as operands.

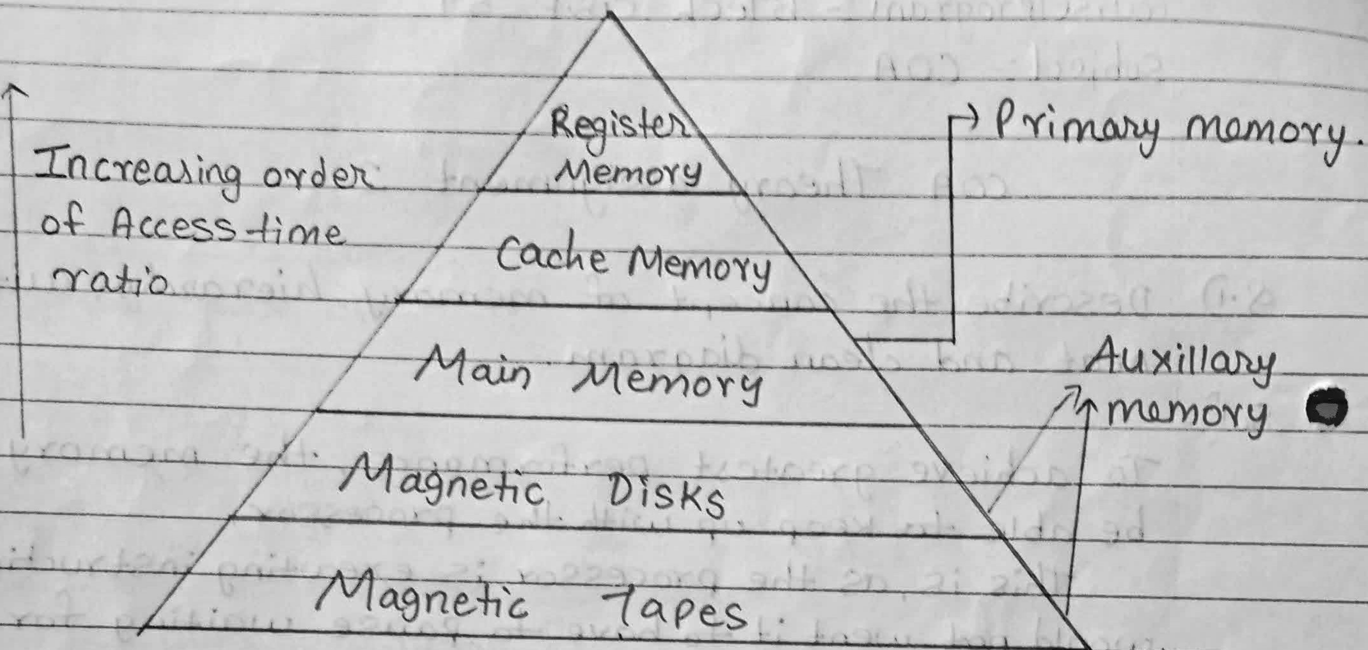
=) Key characteristics of memory are:- capacity, Access time & cost.

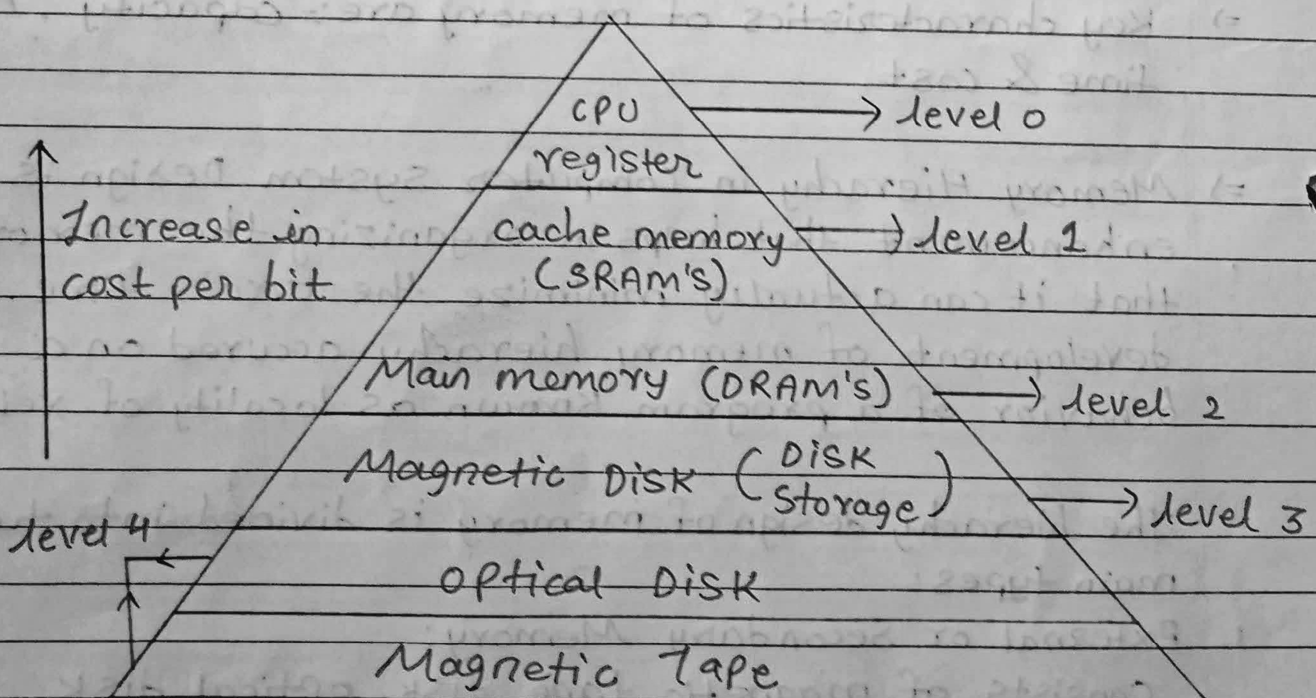
=) Memory Hierachy in computer system Design is an enhancement that helps in organizing the memory so that it can actually minimize the access time. The development of memory hierachy accured on a behavior of a program known as locality of references.

=) The hierachy design of memory is divided into two main types:

1. External or Secondary Memory:
consists of magnetic tape, disk, optical disk (pheripheral storage).
2. Internal or Primary Memory:

consists of CPU, registers, memory and main memory.





Q.2) what are the basic types of Main memory? Explain its advantages.

Sol:- The main memory is the fundamental storage with unit in a computer system and is also called primary memory or internal memory. It is associately large and quick memory and saves programs and information during computer operations. It is divided into two types:

i) Rom ii) RAM

* ROM:- (Read only memory)

It is the type of computer storage containing non-volatile, permanent data that normally can only be read not written to. Rom contains the programming that allows a computer to start up or regenerate each time it turned on. It also performs large I/O tasks and almost every computer incorporates a small amount of Rom that contains the start-up firmware. The boot firmware is called Bios (Basic Input/output system).

=) Types of Read only memory (ROM)

1) PROM (Programmable read only momory)
can be programmed by user and once programmed data and instruction cannot be changed.

2) EPROM (Erasable programmable read only memory).
can be reprogrammed to erase data from it expose it to ultraviolet light. To reprogram it erase all the previous data.

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3). EEPRom (Electrically Erasable Programmable Rom):
Data can be erased by applying an electric field with no need of UV light. can erase only portions.

=) Advantages of ROM
i) NOn-volatile.
Data is not erased when power is off and keeps important updates/programs like BIOS safe.
ii) Security:
Rom is read only so malicious Software/viruses cannot easily alter its contents or affect it.
iii) Booting:-
without RoM, a computer cannot start as BIOS is stored here.
iv) Storage-
Instruction stored in Rom are permanent and unlike RAM, it doesn't need periodic refreshing.

* RAM (Random Access Memory):-
RAM is typically used to refer to memory that is easily read from and written to by the microprocessor. It should be possible to access any address at any time using RAM. It's objective is to store data and applications that are currently in use. The OS gives instructions like when items are to be loaded into RAM, where they are located at or when they are to be removed from RAM.

RAM is further divided into two types:

i) SRAM (Static Random Access Memories):-
RAM's that are made up of circuits and can preserve the information as long as power is supplied is called SRAM. Flip flops from basic elements of SRAM.

ii) DRAM (Dynamic Random access memories):-
A MOS storage cell based on capacitors can be used to replace SRAM cells. such cells cannot preserve the data indefinitely and must be recharged and these cells are used in RAM's and referred to as DRAM.

=) Advantages of RAM:-
1) volatile:
contents are lost when power is off and stores data, instructions and results of currently running programs.
ii) High speed:-
Much faster then secondary memory / storage and allows CPU to quickly read / write data improving overall performance.
iii) Direct-Access:-
Any memory location can be accessed directly and quickly unlike sequential storage (tapes).
iv) Upgradable:-
RAM can usually be increased in most systems, boosting performance without replacing whole computer.

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Q.3) Write difference between RAM & ROM.

Sol^n:-

	RAM	ROM.
1.	Random Access Memory.	Read only Memory.
2.	Volatile	Non-Volatile.
3.	Temporary storage	Permanent storage
4.	Used in normal operations	Used for Startup processes of computer (BIOS).
5.	writing data is faster.	writing data is slower.
6.	Data is lost when power is off.	Data stays safe even when power is off.
7.	Data can easily be changed modified and deleted.	Data cannot be easily changed.
8.	Active during processing	Active during booting.
9.	High storage capacity	low storage capasty.
10.	Types: SRAM, DRAM	Types: PROM, EPROM, EEROM

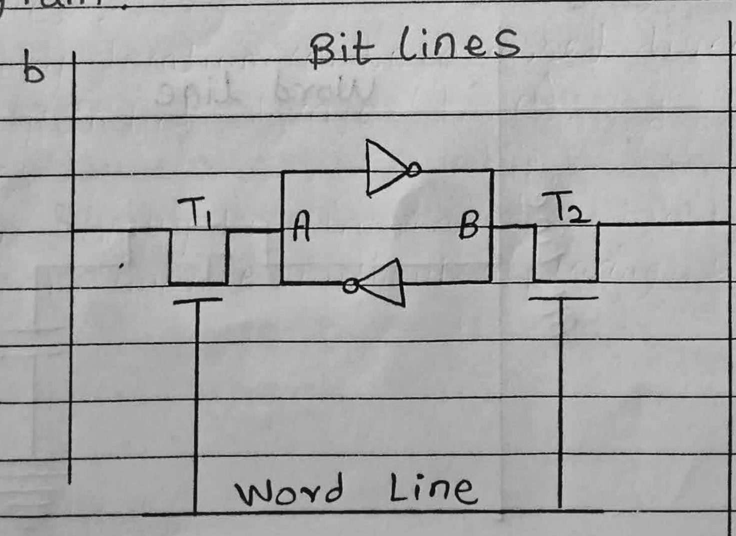
Q.4) Write a short note on

a) SRAM:-

RAM that are made up of circuits and can preserve information as long as power is supplied are reffered to as Static Random Access Memories. An SRAM consists of an array of flip-flops: one for each bit. Because of this, simpler flip-floap circuits, BJT, MOS transistors are used for SRAM.

=> SRAM memory cell:- A latch is formed by two inverters connected. Two transistors T_1 & T_2 are used for connecting the latch with two bit lines. The purpose of transistors is it to act as switches that can be opened or closed under control of word line which is controlled by address decoder. when the word line is at a level, transistors are turned off and latch remains its information.

cell diagram:-



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⇒ for read operation, the word line is activated by the address input to the address. The activated word line closes both the transistors (switches) T_1 & T_2. Then bit value at A & B can transmit to their respective bit lines. The write circuit at end of bit lines sends output to processor.

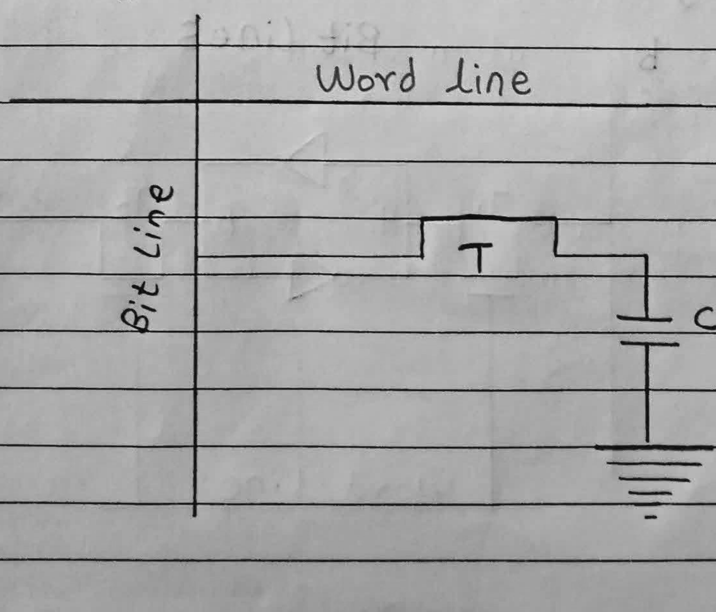
For write operation, address provided to the decoder activates the word line to close both the switches. Then bit value is written into the cell is provided through write circuit and signal are then stored in the cell.

b) DRAM :-

A mos storage cell based on capacitor can be used to replace the SRAM cells. Such a storage cell cannot preserve the charge (data) indefinitely and must be recharged periodically. These cells are called dynatic cells and RAM'S using these cells are called Dynamic RAM'S.

⇒ DRAM memory cell :-

cell diagram :-



=) DRAM uses one transistor and one capacitor each cell where C is the capacitor and R is the transistor Information stored in a DRAM cell is in form of a charge on a capacitor and this charge needs to be periodically recharged.

for storing information in this cell transistor T is turned on and an appropriate voltage is applied to a bit Line. This causes a known amount of charge to be stored in the capacitor. After a transistor is turned off due to the property of a capacitor. it starts to discharge.

for storing information in this cell, transistor T is turned on and an appropriate voltage is applied to a bit line. This causes a known amount of charge to be stored in the capacitor. After a transistor is turned off due to the property of capacitor it starts to discharge

Hence, the information can be read correctly only if it is read before the charge on the capacitors drops below some threshold value.

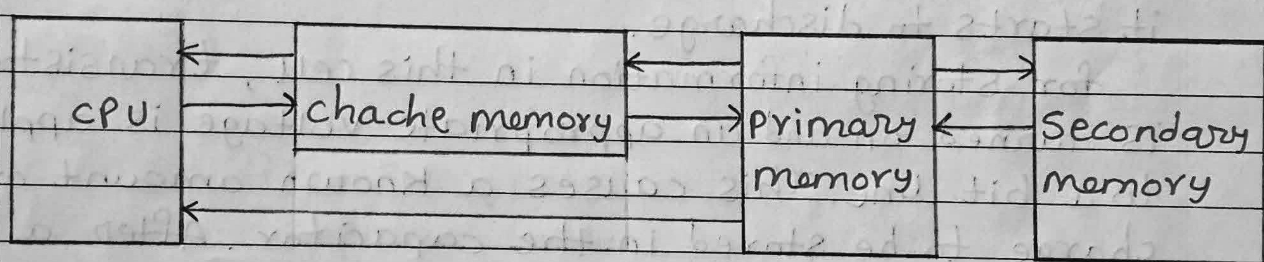
=) There are mainly 5 steps of DRAM:-

1) ADRAM - Asynchronous Dynamic RAM.
2) SDRAM - Synchronous Dynamic RAM.
3) DDR SDRAM - Double data rate synchronous Dynamic RAM
4) RDRAM - Rambus Dynamic RAM.
5) CDRAM - Cache Dynamic RAM

Q.5) What is Cache memory? Explain the levels of a cache memory.

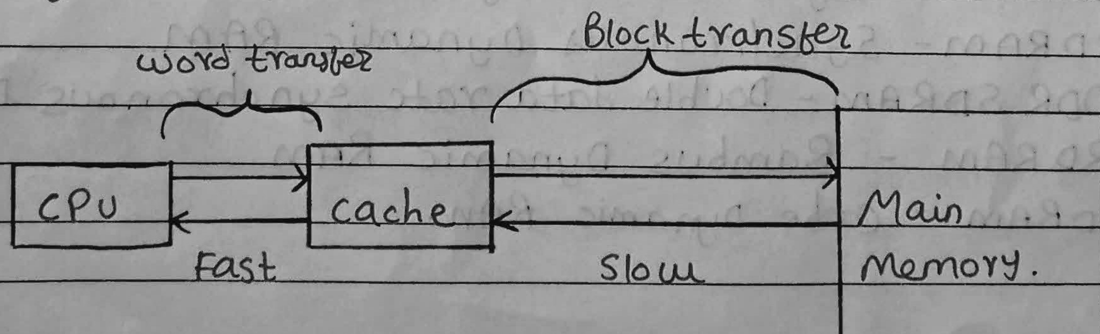
=) cache memory is a high speed memory which is small in size but faster than the main memory (RAM). The CPU can access it more quickly than the primary memory. So it is used to synchronize with high speed CPU and to improve its performance.

It holds frequently requested data and instruction so that they are immediately available to CPU when needed. It is used reduce the average time to access data from main memory.



=) cache makes sure that data is instantly available for CPU whenever the CPU needs this data. If the CPU finds required data or instructions in cache memory, it doesn't need to access to RAM. this by acting as a buffer between RAM & CPU, it speeds up system performance.

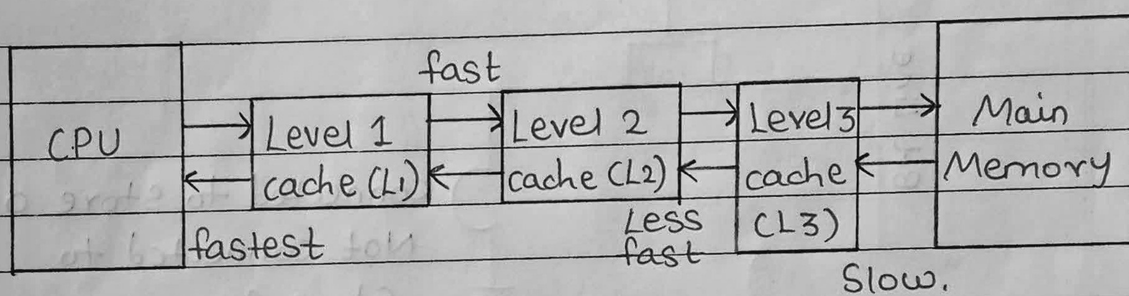
=) Single cache :-



=) when processor attempts to read a word of memory a check is made to determine if the word is in cache. If so, the word is delivered to processor immediately and called ("a cache hit")

If not, a block of main memory consisting of some fixed no. of words is read into cache. then word is delivered as ("a cache miss") to the processor.

=) Three-level cache organization.



=) L1 cache:-
Smallest and fastest cache, located directly on the CPU, stores the most frequently accessed data
(Size:- 16 KB to 128 KB).

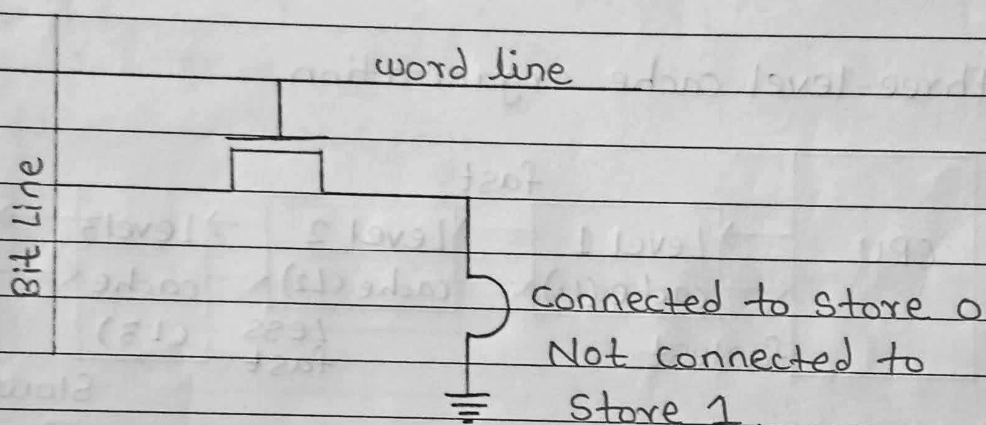
=) L2 cache:-
Largest and slower than L1 but still faster than RAM,
(Size:- 512 KB to 8 MB).

=) L3 cache:-
Even Larger and slower, than L2, shared among multiple CPU cores in modern processors.
(Size:- 30 MB or higher).

Q.6) Explain Rom chip with neat diagram.

Solⁿ:- Rom is a non-volatile memory which means memory cells of this memory chip do not require a power Supply to retain its bit value. As this is a read only memory, the bit values of this memory can only be read and not written or modified.

=) Structure of Rom memory cell:-



=) The bit value of the memory cell is 0 if transistor is at ground level otherwise its 1.

=) The bit line is connected to the power supply via a resistor. To read value of the memory cell the word Line is activated which connects the transistor to the ground.

This drops the voltage of bit line to 0 if the transistor is connected to the ground. If there is no connection between transistor and ground, bit line remains at high voltage indicating 1. The state of the memory cell at the connection to the ground is defined during the chip fabrication.

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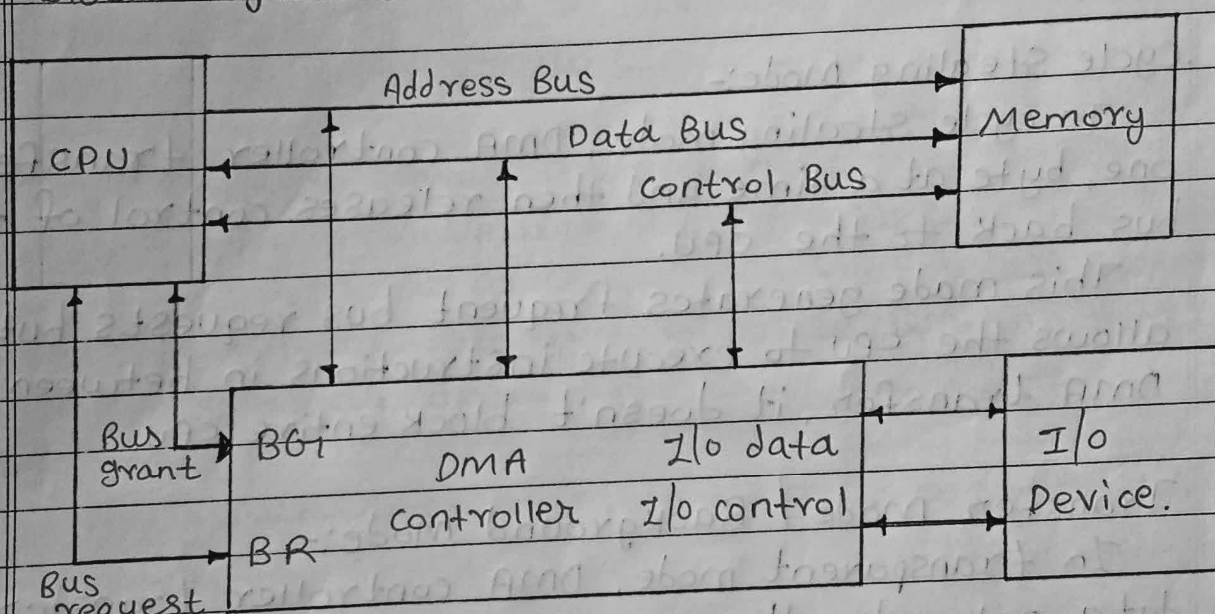
Q.7) With a neat Schematic diagram, Explain about DMA controller and its mode of data transfer.

Sol^n:- Direct Memory Access (DMA) is a technique that allows hardware devices to transfer data directly to/from memory without involving the CPU for every transfer. This reduce CPU overhead and improves system performance

=> Block diagram of DMA controller:-



=> DMA controller is a type of control unit that works as an interface for the data bus and the I/O devices As mentioned, DMA controller has the work of transferring data without intervention of the processors, processors can control the data transfer. DMA controller contains address unit, which generates address and selects a I/O device for the transfer of data.

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⇒ DMA has three data transfer modes:-

1) Burst Data Transfer mode / Block Transfer mode:-
In Burst mode, DMA controller takes full control of the system bus and transfers the entire block of data in one go.
This mode is efficient for large data transfer but can delay CPU operation

2) Cycle Stealing Mode:-
In cycle stealing mode, DMA controller transfer one, byte at a time and then releases control of the bus back to the CPU.
This made generates frequent bus requests but allows the CPU to execute instructions in between DMA transfer, it doesn't block entire CPU.

3) Transfer Mode / Background Mode:-
In transparent mode, DMA controller transfer data only when the CPU is not using system bus. It effectively sneaks in transfers during idle CPU cycle, ensuring CPU is never interrupted.
It works best when CPU performance is critical and some delay in data transfer is acceptable.